Fields and Vector Spaces

Jin-Ren Ryan Wang

June 17, 2026

1 Fields (= Scalar)

1.1 Definition

A set F is called a $field$ if we can define two operators $+$ and $\cdot$ s.t.:

(A0) For any $x, y \in F$, we have $x + y \in F$. (Closure under $+$)

(A1) For any $x, y \in F$, we have $x + y = y + x$. (Community under $+$)

(A2) For any $x, y, z \in F$, we have $(x + y) + z = x + (y + z)$. (Associativity under $+$)

(A3) There $\exists 0 \in F$ s.t. $x + 0 = x$, $\forall x \in F$. (Existence of additive identity)

(A4) $\forall x \in F$, there $\exists x' \in F$ s.t. $x + x' = 0$. (Existence of additive inverse)

(M0) For any $x, y \in F$, we have $x \cdot y \in F$. (Closure under $\cdot$)

(M1) $\forall x, y \in F$, we have $x \cdot y = y \cdot x$. (Community under $\cdot$)

(M2) For any $x, y, z \in F$, we have $(x \cdot y) \cdot z = x \cdot (y \cdot z)$. (Associativity under $\cdot$)

(M3) There $\exists 1 \in F$ s.t. $1 \neq 0$ and $x \cdot 1 = x$, $\forall x \in F$. (Existence of multiplicative identity)

(M4) $\forall x \in F \setminus \{0\}$, there $\exists x''$ s.t. $x \cdot x'' = 1$. (Existence of multiplicative inverse)

(D1) For any $x, y, z \in F$, $x \cdot (y + z) = x \cdot y + x \cdot z$. (Distributivity)

1.2 Examples

1. The sets $\mathbb{R}$, $\mathbb{C}$, $\mathbb{Q}$ are Fields.

2. $M_n(\mathbb{R}) = \{$n by n matrices with entries in $\mathbb{R}\}$ is not a Field. (Mutiplication of matrices is not commutative in general $AB \neq BA$)

2 Vector Space (= spaces that behaves like vectors)

2.1 Definition

Let F be a *field*. A set V is called a *vector space* over F if we can endow with two operations : addition and scalar multiplication by F s.t.:

(A0) $\mathbf{v_1} + \mathbf{v_2} \in V$ for any $\mathbf{v_1}, \mathbf{v_2} \in V$. (Closure under $+$)

(A1) $\mathbf{v_1} + \mathbf{v_2} = \mathbf{v_1} + \mathbf{v_2}$ for any $\mathbf{v_1}, \mathbf{v_2} \in V$. (Commutative under $+$)

(A2) $\mathbf{v_1} + (\mathbf{v_2} + \mathbf{v_3}) = (\mathbf{v_1} + \mathbf{v_2}) + \mathbf{v_3}$ for any $\mathbf{v_1}, \mathbf{v_2}, \mathbf{v_3} \in V$. (Associative under $+$)

(A3) There $\exists \mathbf{0} \in V$ s.t. $\mathbf{v} + \mathbf{0} = \mathbf{v}$, $\forall \mathbf{v} \in V$. (Existence of zero vector)

(A4) $\forall \mathbf{v} \in V$, there $\exists \mathbf{v}' \in V$ s.t. $\mathbf{v} + \mathbf{v}' = \mathbf{0}$. (Existence of additive inverse)

(S0) $\alpha \mathbf{v} \in V$ $\forall \alpha \in F$ and $\mathbf{v} \in V$. (Closure under $\cdot$)

(S1) $\alpha(\mathbf{v_1} + \mathbf{v2}) = \alpha \mathbf{v_1} + \alpha \mathbf{v_2}$ $\forall \alpha \in F$ and $\mathbf{v_1}, \mathbf{v_2} \in V$. (Distributivity)

(S2) $(\alpha + \beta)\mathbf{v} = \alpha \mathbf{v} + \beta \mathbf{v}$ $\forall \alpha, \beta \in F$ and $\mathbf{v} \in V$. (Distributivity)

(S3) $(\alpha\beta)\mathbf{v} = \alpha(\beta \mathbf{v})$ $\forall \alpha, \beta \in F$ and $\mathbf{v} \in V$. (Associativity for scalar multiplication)

(S4) $1\mathbf{v} = \mathbf{v}$, $\forall \mathbf{v} \in V$. (1 as a "scalar identity")

2.2 Examples

1. $V = \mathbb{R}^n$ is a vector space over $\mathbb{R}$

(add on $\mathbb{R}^n$, scalar multiplication by $\mathbb{R}$)

In this case, zero vector $\mathbf{0}$ is given by $(0, 0, \dots, 0)$

In general, let F be a *field* ($\mathbb{R}$ or $\mathbb{Q}$ or $\mathbb{C}$ or $\mathbb{F}_2 \dots$),

$$F^n := \{(x_1, \dots, x_n) : x_i \in F\}$$

2. Let F be a *field*. Define $F[x]$

$$F[x] := \{f(x) = a_0 + a_1 x + \dots + a_n x^n : a_i \in F\}$$

Then $V = F[x]$ (Addition, scalar multiplication by F)

In this case, the zero vector $\mathbf{0}$ is given by "zero polynomial"

3 Basic properties of vector spaces

Theorem 1.4.1. Let V be a vector space over F. Then $\forall \mathbf{v} \in V$, its additive inverse (described in (A4)) $\mathbf{v}'$ is unique. Conventionally, we denote $\mathbf{v}'$ as $-\mathbf{v}$.
Suppose $\mathbf{v}'$ and $\mathbf{v}''$ are two distinct additive inverse of $\mathbf{v}$.

Proof.

$$\begin{aligned}\mathbf{v}'' &= \mathbf{0} + \mathbf{v}'' \\ &= (\mathbf{v}' + \mathbf{v}) + \mathbf{v}'' \\ &= \mathbf{v}' + (\mathbf{v} + \mathbf{v}'') \\ &= \mathbf{v}' + \mathbf{0} \\ &= \mathbf{v}'\end{aligned}$$

Contradiction! Thus additive inverse is unique. □

Theorem 1.4.2. Let V be a vector space over F. Then:

(1) $0 \cdot \mathbf{v} = \mathbf{0}$, $\forall \mathbf{v} \in V$

(2) For any $a \in F$ and $\mathbf{v} \in V$, we have $(-a)\mathbf{v} = -(a\mathbf{v})$

(1)

Proof. Take any $\mathbf{v} \in V$. Let $\mathbf{w} = 0 \cdot \mathbf{v}$ WTS : $\mathbf{w} = \mathbf{0}$

$$\begin{aligned}\mathbf{w} + \mathbf{w} &= 0 \cdot \mathbf{v} + 0 \cdot \mathbf{v} \\ &= (0 + 0) \cdot \mathbf{v} \\ &= 0 \cdot \mathbf{v} \\ &= \mathbf{w}\end{aligned}$$

$$\begin{aligned}\mathbf{w} + \mathbf{w} + -\mathbf{w} &= \mathbf{w} + -\mathbf{w} \\ \implies \mathbf{w} &= \mathbf{0}\end{aligned}$$

□

(2)

Proof. WTS: $(-a)\mathbf{v}$ is the additive inverse of $\alpha\mathbf{v}$

$$\begin{cases}a\mathbf{v} + (-a)\mathbf{v} = (a + (-a))\mathbf{v} \\ \qquad\qquad\quad = 0\mathbf{v} \\ \qquad\qquad\quad = \mathbf{0} \\ a\mathbf{v} + (-a\mathbf{v}) = \mathbf{0}\end{cases}$$

By Uniqueness of additive inverse, we must have

$$(-a)\mathbf{v} = (-a\mathbf{v})$$

□

4 subspaces

Some vector spaces might be 'too big' and be unwieldy to deal with. Therefore, it is often convenient to look at 'smaller vector spaces' inside a given vector space. These are called **subspaces**.

4.1 Definition

Let V be a vector space over F. A subset W of V is called a **subspace of V** if W is also a vector space over F with addition and scalar multiplication inherited from V.

4.2 Theorems

Theorem 1.5.1 Let V be a vector space over F and let $W \subset V$. Then W is a subspace of V iff. the following are satisfied:

- W is not empty;
- W is closed under addition : $\mathbf{w_1} + \mathbf{w_2} \in W$, $\forall \mathbf{w_1}, \mathbf{w_2} \in W$;
- W is closed to under scalar multiplication : $\alpha\mathbf{w} \in W$ $\forall \alpha \in F$ and $\mathbf{w} \in W$.

Proof. ($\Leftarrow$) Suppose $W \subseteq V$ and (1), (2), (3) are satisfied.
Goal: W is itself a vector space. i.e. (A0 $\sim$ A4), (S0 $\sim$ S4) are valid.

(A0) is free of charge because of (2).

(A1) Let $\mathbf{w_1}, \mathbf{w_2} \in W$. Since $W \subseteq V$, we have $\mathbf{w_1}, \mathbf{w_2} \in V$. As V is given, addition on V is commutative

$$\mathbf{w_1} + \mathbf{w_2} = \mathbf{w_2} + \mathbf{w_1}.$$

Hence W inherits commutativity of addition from V.

(A2), (S1)–(S4) Indeed, these axioms are inherited from V.

(S0) is free of charge because of (3).

(A3) Existence of the zero vector.

Since W is nonempty by (1), choose $\mathbf{w} \in W$. By (3),

$$0 \cdot \mathbf{w} \in W.$$

Since $0 \cdot \mathbf{tw} = \mathbf{0}$, we obtain

$$\mathbf{0} \in W.$$

(A4) Existence of additive inverses.

Let $\mathbf{w} \in W$. By (3),

$$(-1)\mathbf{w} \in W.$$

Since $(-1)\mathbf{w} = -\mathbf{w}$,

$$-\mathbf{w} \in W.$$

Therefore W satisfies all vector space axioms, and hence W is a vector space over F.

□

Theorem 1.5.2 Let A be an $m \times n$ matrix. Then the solution space

$$W = \{\mathbf{v} \in \mathbb{R}^n : A\mathbf{v} = \mathbf{0}\}$$

forms a subspace of $\mathbb{R}^n$.

Proof. We verify the three subspace conditions.

1. W is nonempty.

   Since

   $$A\mathbf{0} = \mathbf{0},$$

   we have $\mathbf{0} \in W$.

2. W is closed under addition.

   Let $\mathbf{u}, \mathbf{v} \in W$. Then

   $$A\mathbf{u} = \mathbf{0}, \qquad A\mathbf{v} = \mathbf{0}.$$

   Hence

   $$A(\mathbf{u} + \mathbf{v}) = A\mathbf{u} + A\mathbf{v} = \mathbf{0} + \mathbf{0} = \mathbf{0}.$$

   Therefore $\mathbf{u} + \mathbf{v} \in W$.

3. W is closed under scalar multiplication.

Let $\mathbf{v} \in W$ and $c \in \mathbb{R}$. Since $A\mathbf{v} = \mathbf{0}$,

$$A(c\mathbf{v}) = c(A\mathbf{v}) = c\mathbf{0} = \mathbf{0}.$$

Therefore $c\mathbf{v} \in W$.

Thus W is a subspace of $\mathbb{R}^n$. □

Theorem 1.5.3
If W_1 and W_2 are subspaces of V, then

$$W_1 \cap W_2$$

is also a subspace of V.

Proof. By the Subspace Criterion:

1. Since $\mathbf{0} \in W_1$ and $\mathbf{0} \in W_2$,

$$\mathbf{0} \in W_1 \cap W_2.$$

2. Let $\mathbf{v_1}, \mathbf{v_2} \in W_1 \cap W_2$. Then

$$\mathbf{v_1} + \mathbf{v_2} \in W_1 \qquad \text{and} \qquad \mathbf{v_1} + \mathbf{v_2} \in W_2,$$

so

$$\mathbf{v_1} + \mathbf{v_2} \in W_1 \cap W_2.$$

3. Let $\mathbf{v} \in W_1 \cap W_2$ and $\alpha \in F$. Then

$$\alpha\mathbf{v} \in W_1 \qquad \text{and} \qquad \alpha\mathbf{v} \in W_2,$$

so

$$\alpha\mathbf{v} \in W_1 \cap W_2.$$

Hence $W_1 \cap W_2$ is a subspace of V. □

4.3 Example of subspaces

4.3.1 Subspaces of $\mathbb{R}^n$

Show that

$$W = \{(x, y) \in \mathbb{R}^2 : x + 2y = 0\}$$

is a subspace of $\mathbb{R}^2$.

Proof. By the Subspace Criterion, it suffices to verify that W is nonempty, closed under addition, and closed under scalar multiplication.

1. $(0, 0) \in W$ since $0 + 2 \cdot 0 = 0$.
2. Let $\mathbf{w_1} = (a, b), \mathbf{w_2} = (c, d) \in W$. Then

$$a + 2b = 0, \qquad c + 2d = 0.$$

   Hence

$$(a + c) + 2(b + d) = (a + 2b) + (c + 2d) = 0.$$

   Therefore $\mathbf{w_1} + \mathbf{w_2} \in W$.

3. Let $\mathbf{w} = (a, b) \in W$ and $r \in \mathbb{R}$. Then

$$ra + 2(rb) = r(a + 2b) = 0.$$

   Therefore $r\mathbf{w} \in W$.

Hence W is a subspace of $\mathbb{R}^2$. □

4.3.2 Subspaces of polynomial space

$F[x]_{\leq n}$ is a subspace of $F[x]$.

Proof. By the Subspace Criterion, it suffices to verify that $F[x]_{\leq n}$ is nonempty, closed under addition, and closed under scalar multiplication.

1. The zero polynomial belongs to $F[x]_{\leq n}$, so

$$0 \in F[x]_{\leq n}.$$

2. Let $f, g \in F[x]_{\leq n}$. Then

$$\deg(f) \leq n, \qquad \deg(g) \leq n.$$

Hence

$$\deg(f + g) \leq \max\{\deg(f), \deg(g)\} \leq n.$$

Therefore

$$f + g \in F[x]_{\leq n}.$$

3. Let $f \in F[x]_{\leq n}$ and $c \in F$. Then

$$\deg(cf) \leq \deg(f) \leq n.$$

Therefore

$$cf \in F[x]_{\leq n}.$$

Hence $F[x]_{\leq n}$ is a subspace of $F[x]$. $\square$

4.3.3 Subspaces of Function Spaces, take differentiable function for example

Let

$$C(D) = \{f : D \to \mathbb{R} : f \text{ is continuous}\}.$$

Define

$$D(D) = \{f : D \to \mathbb{R} : f \text{ is differentiable}\}.$$

$D(D)$ is a subspace of $C(D)$.

Proof. By the Subspace Criterion, it suffices to verify the following.

1. The zero function belongs to $D(D)$.
2. Let $f, g \in D(D)$. Since the sum of two differentiable functions is differentiable,

$$f + g \in D(D).$$

3. Let $f \in D(D)$ and $\alpha \in \mathbb{R}$. Since a scalar multiple of a differentiable function is differentiable,

$$\alpha f \in D(D).$$

Therefore $D(D)$ is a subspace of $C(D)$. $\square$

4.3.4 Subspaces of matrix space, take symmetric matrix for example

Recall that the transpose of a matrix $A = (a_{ij})$ is

$$A^T = (a_{ji}).$$

Define

$$\mathrm{Sym}_n(F) = \{A \in M_n(F) : A^T = A\}.$$

The matrices in $\mathrm{Sym}_n(F)$ are called symmetric matrices.

$$\mathrm{Sym}_n(F) \text{ is a subspace of } M_n(F).$$

Proof. By the Subspace Criterion, it suffices to verify that $\mathrm{Sym}_n(F)$ is nonempty, closed under addition, and closed under scalar multiplication.

1. Since

$$0^T = 0,$$

the zero matrix belongs to $\mathrm{Sym}_n(F)$.
2. Let $A, B \in \mathrm{Sym}_n(F)$. Then

$$A^T = A, \qquad B^T = B.$$

Hence

$$(A+B)^T = A^T + B^T = A + B,$$

so $A + B \in \mathrm{Sym}_n(F)$.
3. Let $A \in \mathrm{Sym}_n(F)$ and $c \in F$. Then

$$(cA)^T = cA^T = cA,$$

so $cA \in \mathrm{Sym}_n(F)$.

Hence $\mathrm{Sym}_n(F)$ is a subspace of $M_n(F)$.

□